Max Vilgalys

max.vilgalys@gmail.com | (425) 449-9464 | vilgalys.github.io

Education

Massachusetts Institute of Technology, Ph.D., Social and Engineering Systems — Sep. 2022
Dissertation: Essays on Measuring Climate Change Damages and Adaptation
Committee: Jing Li, Namrata Kala, Whitney Newey
Coursework includes: Statistics, microeconomics, industrial organization, optimization, experiment design, energy and environmental economics, nonlinear econometrics

Stanford University, B.S., Electrical Engineering — June 2017
Coursework includes: Systems programming, probability and statistics, machine learning, energy systems

Experience

Supervising Analyst, New York City Office of Management and Budget — August 2023-Present
- Researched and testified on cost of capital in utility rate cases on behalf of New York City
- Studied emissions reductions from local building laws with novel financial decision model
- Helped secure local law extending affordable housing subsidy program
- Worked with U.S. EPA to forecast New York City's energy use and emissions (COMET-NYC, 2025)
- Performed analysis, developed data visualizations, researched, and contributed to New York City's first (2024) and second (2025) climate budget publications
- Presented research to NYC Budget Director, City Council, city agencies, and external parties

Economist, Eastern Research Group — Nov. 2022-Aug. 2023
- Researched and wrote chapter on risks from extreme temperature in Massachusetts
- Developed procedure to estimate uncertainty in US greenhouse gas emissions inventory
- Conducted cost-benefit analysis and data analysis to support climate and environmental projects
- Designed natural language processing approach to automatically classify workplace injury reports

Graduate Student Researcher, MIT — Sep. 2017-Sep. 2022
- Conducted independent research projects on measuring adaptation to climate change in agriculture
- Developed and implemented methodologies, using machine learning and structural econometrics
- Analyzed large spatial and time series datasets of weather, crop yields, and wildfire reports
- Assisted Professors Nikhil Agarwal, Jing Li, and Dava Newman in research projects; conducted geospatial data analysis, tested and implemented structural models, developed data visualizations

Teaching, New York University, MIT, and MIT EdX
- Instructor: Innovative City Governance (NYU CUSP, Win. 2025, Fall 2026)
- TA: Science, Technology and Public Policy (MIT, Jan.-May 2022), Statistics, Computation and Applications (MIT EdX online course, Sep.-Dec. 2021; MIT Jan.-May 2021), Probability (MIT EdX online course, Sep.-Dec. 2020)

Working Papers: Affordability and equity in New York City's building performance standard (2025), Estimating Continuous Treatment Effects in Panel Data using Machine Learning with a Climate Application (2023), Equity and Adaptation to Wildfire Risk: Evidence from California Public Safety Power Shutoffs (2023), A Machine Learning Approach to Measuring Climate Adaptation (2023)
Fellowships and Awards: NYC Hayes Innovation Prize Finalist (2024), MIT IDSS TA Award (2022), U.C. Berkeley/Sloan Summer School in Environmental and Energy Economics Diversity Fellowship (2020), MIT Exxon Mobil Energy Fellow (2019-2020), MIT Presidential Fellow (2017-2018)
Programming: Python, MATLAB, Julia, Stata, R, C#/C/C++, SQL
Languages: English (native), German (conversational)
Citizenship U.S., Germany